

2 Goal and Area of Application

Mass spectrometry-based chemical proteomics is a cornerstone technology for elucidating the molecular targets and mechanisms of action (MoA) of small molecules, biologics, and other therapeutic modalities within the REMEDI4ALL drug-repurposing framework. Similar to phenotypic profiling approaches such as Cell Painting, chemical proteomics provides an unbiased, system-level readout of how compounds perturb biological systems, enabling robust insights into target engagement, protein network rewiring, and downstream pathway alterations.

Drug repurposing efforts rely on understanding *why* and *how* a compound exerts its biological effects in a new therapeutic context. However, many drugs entering clinical evaluation fail because the initially proposed MoA is incomplete or incorrect. Since most drug targets are proteins, and proteomes differ substantially across tissues and disease states, accurate target deconvolution (TD) and MoA elucidation in relevant models are critical for successful repositioning strategies.

Chemical proteomics enables the identification of proteins whose abundance, solubility, conformation, interaction landscape, or post-translational modifications change upon drug exposure – both rapidly (minutes to a couple of hours) and at later regulatory time points (several hours or days). These protein-level responses can reveal:

- Direct molecular targets
- Functionally relevant off-targets or liabilities
- Protein complexes and protein networks affected by drug binding
- Early mechanistic events preceding transcriptional or phenotypic responses
- Pathways and networks underlying therapeutic or adverse effects
- Proteins reduction or oxidation on cysteines during redox drug mechanisms (peptide level)
- Binding site of drugs on protein targets and allosteric rearrangements

The platform combines orthogonal MS-based strategies, including PISA, expression proteomics, RedOx proteomics and affinity-based chemoproteomics, to build a multidimensional, proteome-wide, molecular fingerprint of compound action. These complementary approaches offer deep proteome coverage, high throughput, and the statistical robustness needed for objective TD and MoA analyses in drug-repurposing projects. In addition to those proteome-wide strategies, for target-based studies, for example after target deconvolution, HDX-MS analysis can also be carried as structural proteomics analysis on purified proteins, validating direct target engagement in solution, mapping and characterizing the binding site and allosteric protein rearrangement upon drug-binding.

Applications include:

- MoA characterization of hits from phenotypic screens
- Validation of hypotheses from computational or image-based profiling
- Comparison of repurposing candidates with mechanistically annotated reference compounds
- Investigation of polypharmacology or off-target toxicity
- Mechanistic characterization in primary cells or low-input precision-medicine models

Ultimately, mass spectrometry-based chemical proteomics provides an unbiased, scalable, and mechanistically rich readout that complements phenotypic assays and accelerates the development of evidence-based, clinically relevant repurposing hypotheses within REMEDI4ALL.

3 Definitions and Abbreviations

KI	Karolinska Institutet
ChemProt	C hemical P roteomics Unit, MBB-Department, Karolinska Institutet
TD	T arget D econvolution
MoA	M echanism of A ction
PISA assay	P roteome I ntegral S olubility A lteration assay
FITeXP	F unctional I dentification of T argets by E xpression P roteomics
MS	M ass S pectrometry
MS/MS	T andem M ass S pectrometry
LC	L iquid C hromatography
TMT	T andem M ass T ag
HDX-MS	H ydrogen / D euterium eX change M ass S pectrometry
FT-IsoR-MS	F ourier T ransform I sotopic R atio M ass S pectrometry
RedOx	R eduction / O xidation balance
PTM	P ost T ranslation M odification
HOLSER	H igh-rati O partial proteolysi S with carri E R proteome
PDB	P rotein D ata B ank

4 Method

4.1 General

Mass spectrometry-based chemical proteomics provides a powerful and unbiased strategy for target deconvolution (TD) and to elucidate mechanisms of action (MoA) for drugs and repurposing candidates. As in other phenotypic or profiling approaches, understanding how a compound perturbs cellular systems at the protein level is essential, especially since many drugs fail in clinical development due to incorrect or incomplete MoA assignments. Because most drug targets are proteins and proteomes differ substantially between cell types,

proteome-wide measurements in relevant biological models are critical for drug-repurposing efforts.

Chemical proteomics enables the measurement of direct and indirect protein-level effects of a drug in living cells or lysates, capturing both immediate protein alterations and downstream effects on the proteome and cellular regulatory responses. These may include:

- Proteome-wide profiling of protein solubility shifts in cells or native lysates to determine direct biophysical changes upon drug–target interaction
- Conformational and changes of protein accessibility in solution to proteolytic enzymes within native proteomes
- Changes in total protein abundance as time resolved mechanistic proteome regulation
- Modulation of protein complexes or networks in early phases preceding gene-expression changes or in later stages downstream to functional regulation of cell proteome
- Post-translational modification and proteome regulation
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Together, these measurements allow researchers to build a multidimensional mechanistic fingerprint that clarifies how a compound acts, whether additional off-targets contribute to its phenotype, and how cellular pathways and networks respond to treatment.

To achieve this, a suite of orthogonal and integrable proteomics approaches is used:

- PISA (Proteome Integral Solubility Alteration) for proteome-wide, high-throughput and high multiplexing proteome profiling for identifying proteins whose solubility shifts immediately upon target engagement in cells and cell lysates
- Expression/global proteomics to quantify protein abundance changes following time resolved drug treatments
- Affinity/activity-based chemoproteomics to characterize interactomes using suitable engineered probes
- RedOx proteomics to identify proteins and their cysteine containing peptides affected by oxidative mechanisms that may contribute to MoA
- PTM-focused proteomics (e.g., phosphoproteomics) to resolve signaling-level perturbations and post-translational proteome regulation
- HDX-MS to validate drug-target interactions in solution, map interaction sites and characterize structural and conformational drug-induced changes in purified target proteins upon binding

Each of these methods provides complementary information, and they can be combined to overcome the limitations of any single technique. Together they allow detection of thousands of confidently quantified proteins, enabling robust statistical comparisons across treatments, concentrations, and biological replicates.

Chemical proteomics experiments begin with a carefully designed study plan with experimental design exploiting the multiplexing of comparative parameters relevant to MoA

characterization, such as different treatment conditions and drug analogues, drug concentrations, sample scale and nature, and expected cellular responses. For meaningful MoA interpretation, compatibility with phenotypic assays is maintained, and a statistically relevant number of biological replicates in the biological models relevant to the intended repurposing, with appropriate controls are included. Overall, mass spectrometry-based approaches provide a scalable, deep, and unbiased platform for Target deconvolution and MoA elucidation. Their ability to detect early target engagement events and integrate with protein network and pathway analyses makes them central to the molecular characterization of compounds within REMEDI4ALL drug-repurposing projects.

4.2 Procedures

4.2.1 Procedure for TD and MoA elucidation

Experimental design, selection and integration of proteomics approaches

The first step is to identify the main research questions about the drug, or drug set - on which select and design the main approach or the integration of orthogonal ones –, and considering the following:

- the existing knowledge and preliminary data (basic research and clinical studies) related to the drug and the potential clinical settings of application;
- the adoption of one or multiple biological models to study the new indications, such as cell lines or primary cultures of interest;
- the relevance and availability of using cell cultures with different sensitivity levels to the drug of interest;
- the available amount, nature and heterogeneity of the biological samples;
- the number of drugs and treatments to be studied in synergy or in comparison for being able to draw specific questions to the resulting proteomic dataset;
- the time of analysis after treatment if specific drug metabolism or induced intracellular processes can be relevant for MoA, such as, for example, induction or redox changes;
- IC₅₀ / EC₅₀ measurements in phenotype-based assays (e.g. cell viability assays) in a time-resolved manner, for example at 24h, 48h or longer exposure times (48h is a usually a good time point as the proteome response in living cells reaches a sort of end point for its regulation in response to the drug, while an average cell proliferation rate should not affect a specific selection of cells);
- the possibility to use drugs similar in structure or different drugs with similar phenotypic effects, to sort out similarities and differences in the protein datasets intersection;
- the possibility to use a drug analogue which is close in structure but biologically not active or much less active on the same cell cultures used as biological model;

- the use of multiple concentrations and times of treatment if this does not affect the number of drugs needed to be studied in comparison in one assay;
- the relevance of treating lysates / extracts in addition to proteomics on living cells.

After completion of the experimental design, TD and MoA studies are carried out through coordinated workflows from the reception of cells (or cell lysate) and drugs of interest, through cell culture, treatment, proteomics processing, LC-MS, database search, RAW data analysis, data visualization and interpretation. To preserve the biological model(s), cell cultures are handled in conformity to the modalities used for compound testing in phenotypic assays.

General aspects for different proteomics approaches in TD and MoA elucidation

As most of the research questions require a quantitative answer, quantitative proteomics approaches need to be used. Deep proteome coverage is required for proteomics analysis for unbiased TD and MoA elucidation, therefore nanoflow LC connected to electrospray ionization and MS/MS analysis is used.

Two major modalities of MS/MS acquisition can be used with modern MS instrumentation.

- 1) In data dependent acquisition, the precursor ions are selected starting from most abundant ions detected of a full MS spectrum while eluted from in source heated LC columns, isolated and fragmented for high resolution MS/MS analysis, with temporary exclusion after analysis allowing the lower abundant ions to be analysed.
- 2) In data independent acquisition, precursor ions are isolated into pre-defined isolation windows and fragmented, then all fragmented ions in each window are analysed by a high-resolution MS/MS.

Both methods can provide deep proteome coverage, with the second method being the most recent and suitable for deep label-free quantitative proteomics analysis in single and consecutive runs of sets of samples. However, the first method is preferred because it can be combined with TMT technology for quantification of isobaric ion reporter allowing to multiplex up to 18 biological samples and peptides can be extensively fractionated upstream LC-MS analysis for in-depth proteome profiling. After protein extraction of each sample, total protein amount is quantified and the same amount is precipitated, reduced, alkylated and digested into peptides before to be labelled by a different TMT ion reporter. The different peptides mixtures after labelling are combined in one multiplexed sample and, every time a parental ion is fragmented and analysed, the different reporter ions are detached and quantified in the same MS spectrum simultaneously, enabling relative quantification over control samples in the same multiplex. This technology, combined to fractionation of the multiplexed peptide sample via high pH, reversed-phased capillary LC prior to LC-MS/MS still provides the deepest proteome coverage and the most robust quantification across different experimental conditions in comparison to other methodologies.

The TMT-based workflow described above, peptide fractionation and data dependent LC-MS/MS acquisition, is the one applied for all the different approaches for TD and MoA described in the next sessions.

Stable isotope labelling using amino acids in cell culture can also be used for specific questions (*e.g.* protein turnover), however the multiplexing capacity is not sufficient for unbiased TD and MoA elucidation.

Proteomics approaches / tools for TD and MoA

1 The PISA assay identifies proteins - among more than 7500 or 8000 proteins confidently identified and quantified ⁶ with at least two unique peptides across all biological replicates used for each assay (up to 18)- which soluble amount changes relatively to controls upon treatment across a gradient of a solubility perturbation (protein precipitation) agent (*e.g.*, temperature, ions, solvent). Temperature is most frequently used as more practical to be quenched relatively to chemical agents, but solvent-based PISA can be performed with similar outcomes and performance of temperature-based PISA. When a drug enters cells, the engagement of the target produces a biophysical alteration of the last, read by solubility change. PISA is carried out generally with a short incubation time, generally up to 60 minutes of drug treatment, when gene expression responses to the treatment are not yet affecting the total protein amount in cells. Other proteins, like off-targets (either functional or not affecting the phenotypic outcome), proteins complexed to the targets, as well as proteins early modified in their complex's organization in the proteome, or structurally modified by fast post-translational modification in response to drug-target engagement, can also be altered in solubility. After the application of the solubility alteration gradient on equal cell sample aliquots, cell proteomes are extracted avoiding strong detergents, all the aliquots of each samples are pooled and ultracentrifuged at more than 100 000 $\times g$ to isolate the soluble proteome fraction of each sample. Then the TMT-based workflow described above for quantitative proteomics is applied for protein identification and quantification. Both positive and negative soluble amount shifts across the whole gradient of alteration of solubility, of each condition compared to controls are considered, as the total solubility resultant in the complexity of the proteome matrix could be either positive or negative. The size of solubility shift relative to control is a major factor, but not the only one, as the total amount of biophysical stability in solution of each protein can depend more intrinsically on the nature of the target (cytosolic, or membrane, nuclear) rather than the drug. The use of at least three or four biological replicates per condition within the multiplexed analysis provides the statistical confidence of the solubility alteration event. Therefore, both the transformed and absolute values of the soluble amount ratios and p-Values of each treatment, relative to controls, are used to score and rank the proteins most shifting on solubility. The top-ranked proteins include the target, and the final target deconvolution is given by the correlation and intersection of the datasets produced by conditions applied as per the initial experimental design within the same

PISA assay. Protein network analysis can identify known protein complexes involvement initiating the drug MoA.

This method has major advantages, such as: no need of chemical probe engineering; can be carried out on living cells or lysates; deep, high throughput and multiplex proteomics analysis, with a well-sustainable use of statistically relevant number of biological replicates and low number of cells; no need and lack of biases due to curve fitting manipulation of the protein quantification performed by LC-MS/MS; very versatile in experimental design with room for internal orthogonal integration of other chemoproteomics approaches.

PISA can be carried out with adapted workflows for bacteria and miniaturized for mammalian cells culture models with a low cell number, such as primary cultures and iPS-derived cells. PISA can be carried out incubating the drug in protein extracts for identification of potential protein targets in protein extracts, in addition or integration with PISA in living cells. Additionally, PISA in cell lysates is optimized for identification of targets and off-targets of therapeutic antibodies used as drugs, with evaluation of their specificity / selectivity. Most targets change their solubility point and therefore soluble amounts in PISA in a reproducible manner upon ligand binding, because of direct binding or relative association to other molecular complexes or networks, however there might be some cases in which protein targets do not show significant alteration of solubility upon ligand interaction, due to other major factors, ligand independent, strongly contributing to their solubility and practically making solubility effects of ligand binding neglectable. This calls for complementary proteomics approaches or cross-validation strategies.

2 Identification of the interactome and protein complexes after affinity/activity-based chemoproteomics with engineered probes, also cooperating with Chemical Biology Consortium Sweden for chemical characterization, design and engineering. This method requires probe engineering and robust pull-down or immunoprecipitation approaches to minimize background interference and variability across samples during the enrichment phase. Elution fractions can be analysed quantitatively, but with a potentially high variability because of the additional sample manipulation in the enrichment phases ⁷.

3 Expression / global proteomics at different time points of treatment to evaluate the differential protein amounts at the proteome scale as proteome regulation response of cells upon treatment. This approach uses at least triplicates and can be integrated with PISA (PISA-Express) in one single assay for measuring total and soluble protein amount changes upon long term treatments as well as for elucidation of the drug MoA evolution in cells ⁸.

4 Expression proteomics for drug-specific proteome regulation responses using multiple drugs in a FITeXP mode, or involving the ProtarGetMiner proteome signatures' database for anticancer molecules ^{9,10}. This approach is designed using multiple treatments producing similar cellular phenotypes at 48h with their IC50, where the different drug datasets of

proteome regulation are analysed together to sort out the specific protein regulation of a drug from the common regulation of general processes leading to the phenotypic effect. For anticancer drugs, any expression proteomics datasets of any drug treatment can be also analysed in FITeXP mode using the analysis tool ProTargetMiner (<https://protargetminer.genexplain.com>) allowing the comparison of 56 anticancer drugs in one or three cell lines, to elucidate common targets or MoA with other anticancer drugs.

5 High throughput RedOx proteomics (recently developed) for changes in the Red-Ox state of proteins. Reactive oxygen species (ROS) are metabolic or signalling molecules often involved in drug MoA that exert their effects by oxidizing protein amino acid residues, particularly cysteines (Cys), which can lead to modulation of protein activity or dysfunction. This method makes use of the proteome-wide and deep proteomics workflow described in the previous session using TMT. The workflow is modified by splitting each sample in two and proceeding with two parallel processing involving alkylation before and after reduction. The RedOx state is analysed at specific timepoints of drug treatment and requires normalization over the total peptide/protein amount changes, including RedOx and Expression proteomics in one (REX) for the RedOx timepoint of analysis (*e.g.*, 24h). The quantitative analysis is carried at the peptide level measuring the RedOx change in proteins and peptides containing Cysteines (cysteinome) overcoming the bottlenecks of previous methods on the yield of concurrent identification of reduced and oxidized forms of the cysteine-containing peptides in all samples. This method is also integrable into PISA in one assay determining for one or two drug protein solubility alteration, expression level changes and RedOx state variation of proteins at the proteome level (PISA-REX) ¹¹.

6 Analysis of PTMs involved in the MoA (*e.g.*, Phosphoproteomics, etc.) ¹². This approach makes use of a similar workflow based on TMT analysis described before. However, it includes enrichment steps for the peptides containing the desired PTMs. The TMT labelling step is carried out before enrichment, which is performed after combination of all biological replicates / conditions of the designed experiment.

7 FT-IsoR-MS, a new method capable of measuring the differential isotope composition of proteomes and proteins analysing different types of molecular isotopes, with current applications in biomedicine, environmental biology, and archaeology ¹³. In drug repurposing can be used for measuring the incorporation of specific isotopes in specific tissues of interest in animal models, or in enriched sub-proteomes or selected proteins as effect of a drug treatment of drug and evaluation of an altered metabolism.

8 HOLSER, High-ratiO partial proteolysiS with carriER proteome (HOLSER), a new method enabling global structure profiling and site-resolved elucidation of ligand–protein interactions. Partial proteolysis-based proteome profiling methods, such as peptide-centric local stability assay (PELSA) offer peptide-level resolution of ligand-induced conformational changes but are limited by modest proteome coverage and depth, as well as high variability. HOLSER is an

optimized proteome-profiling workflow designed to measure structural changes with high depth and reproducibility. It integrates aggressive yet controlled proteolysis with extended digestion to reduce variability, and uses TMT-based multiplexing that includes full digests to boost quantitative precision and proteome coverage. This combination enables HOLSER to detect domain-specific stabilization, identify ligand-contact peptides, and analyze conformational responses across entire proteomes as well as within specific protein complexes or binding sites.

Data analysis of single and integrated proteomics strategies

Acquired LC-MS/MS data is searched against proper proteome database of reference using mostly licensed software, *e.g.*: Proteome Discoverer 3.1 with SEQUEST, MASCOT or ANDROMEDA search engines, PEAKS Studio 10.6, Biopharma Finder, but also the free software MaxQuant can be used. Other associated or internally developed tools for protein identification and quantification can be also applied.

For data analysis and visualization, different scripts and pipelines based on R or Python are applied to produce an interactive HTML report, including plots and tables for each drug treatments, as well as comparative and correlation analysis across the different experimental conditions or integrated experiments, for gaining TD and MoA insights. The analysis can be carried out carefully also through Excel or similar tools of calculation. Each quantitative value is normalized on the total protein amount detected for each sample in all fractions, followed by normalization on the control values. Ratio of the soluble amount, p-Values and their Log transformed correspondent are generated. In Expression proteomics the size effect of the protein amount changes is considered while in PISA the ranking is provided by a score composed by the size effect of the soluble amount variation and its statistical significance across replicates.

A new tool for data visualization is under technology development at SciLifeLab for projects including multidimensional -omics data or projects involving different Units at the Chemical Biology and Genome Engineering platform (CBGE), by cooperation of the NBIS (National Bioinformatics Infrastructure Sweden) and CBGE.

Statistical analysis is performed using T-test, False Discovery Rate or Bonferroni corrections, and permutation analysis.

Supervised and unsupervised analyses of the proteomics datasets, such as Principal component analysis and Partial Least Squares Discriminant Analysis, are usually carried out using the SIMCA software.

Publication-quality figures and plots are drawn by Prism (GraphPad). Protein network analysis is carried out by STRING analysis (<https://string-db.org/>) using the free online tool or associated R scripts. Ingenuity Pathway analysis software and other pathway analysis software can also be used.

MS-based proteomics data are deposited upon request onto the community data repository PRoteomics IDentifications database, hosted by EMBL-EBI, <https://www.ebi.ac.uk/pride/>).

4.2.2 Procedure for characterization of the drug-target interaction

Experimental design and planning

The experiment is mostly designed based on the research data available on the protein target regarding structure-related information.

HDX-MS is performed to map binding sites and conformational protein changes upon protein interaction with any ligand. For drug repurposing aims the ligand can be a drug in the form of a small molecule or another protein such as an antibody for epitope mapping. In the latter case also paratope mapping can be provided by HDX-MS.

With HDX-MS a purified protein is analysed in solution for gathering information on conformational and structural changes when the protein is engaged in drug binding, also verifying the direct target engagement in solution. In HDX-MS the aim is to use MS-based proteomics tools to map the sequence of the protein at its native conformation in solution and measure, between the protein state of free and drug-engaged protein, the exchange of the amide hydrogen of the peptide bonds with deuterated buffer, which can exchange in the timescale of seconds to minutes and hours dependently on their solvent accessibility in the conformation. This provides information on the structural changes of a protein in solution even when other protein structure methods fail, with a much shorter analysis than other methods (a few weeks to obtain results). HDX-MS provides valuable insights on the protein in solution with and without the binding also in combination of structural information acquired with other techniques.

General aspects of HDX-MS methodology

The tested protein is deuterated at different time points, three or more, generally from 30 seconds (or shorter if needed) up to 60 min, in triplicates, to measure the H/D exchange dynamics of each peptide over time comparing the bound and unbound states of the investigated protein in solution. The exchange is then quenched at low temperature and low pH then the protein is injected into a pepsin column for protein digestion into peptides at those conditions. Digested peptides are trapped in-line on a column and then eluted for LC-MS analysis. Peptides are identified and their incorporation of deuterium is measured in all replicates of all time points of the two protein states.

The whole process is very sensitive to precision in time and temperature, also to reduce D / H back-exchange, and is carried out using an automated liquid handler specific for HDX-MS, allowing to precisely control the full in-line workflow from the deuteration up to LC-MS.

There is no limitation on protein size, the success rate depends mostly on the protein solubility and digestion rate in a detergent-free buffer, as detergents cannot be removed before MS. Buffer exchange must be carried out and protein concentration evaluated if the initial buffer contains detergents, glycerol, or MS-incompatible chemicals.

A feasibility study on the free protein form in solution is due as first thing, to optimize experimental conditions and ensure that most of the protein sequence is well covered by the peptides digested and analysed, then the full experiment is carried out for all time points and replicates. The bound form of the drug-protein complex is obtained using molar excess of the drug, with predictive molar excess depending on expected affinity values, generally from 3 to 10 molar excess.

The data is analysed with a licensed software calculating the deuterium incorporation and the differences H/D exchange differences and producing heatmaps on the sequence that can be over imposed on the deposited PDB structure used as model.

HDX-MS experimental guidelines

Required information before to start the experiments:

- Exact sequence of the protein and its reference in the proteome database
- Protein amount concentration of the sample and level of purity. At least 300 μg protein, divided in 5-10 aliquots is needed for the interaction study with one ligand (drug)
- The solubility of the protein in the final buffer, ideally soluble up to at least 1mg/ml
- The buffer composition of the protein suspension. If MS-incompatible chemicals are present, buffer exchange is needed checking protein concentration and solubility limits
- The PDB entry number, checking the difference with the real protein sequence
- The binding affinity between the ligand and the protein, if known or predicted
- Incubation condition for any binding affinity assay performed, such as concentration of the ligand and the protein, temperature, incubation time, *etc.*
- The maximum concentration of the drug ligand in organic solvent, for example DMSO), as the saturation needs to be achieved minimizing DMSO amount
- Samples (free and drug-bound) are analysed in an automated HDX-MS system (HTC PAL) in which samples are automatically labelled, quenched, digested, cleaned, and separated before MS measurement

Feasibility test:

- the aim is to optimize the quenching buffer, digestion temperature, amount of protein, deuteration temperature and deuteration time points.

- The feasibility test represents an important and long part of the experiment, basically using only the free protein form, in order to set the optimized conditions and must provide a protein sequence coverage by the identified peptides of at least 75%
- The quenching buffer contains denaturants and acidic conditions that can be tuned to improve the digestion of the protein by the pepsin column at low temperature (5-10 °C)
- The deuteration reaction is performed in automated manner by mixing 4 µL of sample with 36 µL of deuterated buffer.
- The deuteration temperature can be set at 20 °C. However, if the deuteration is too fast, the temperature can be lowered
- Deuteration time points can be chosen as 30s, 5min, 20min, 60min with single injection, if the deuteration percentage is still low after 60min, longer time should be performed. If the deuteration percentage at 20min is like 60min, no longer time is needed
- After incubation, the reaction is quenched by decreasing the pH to ~ 2.3 and the temperature to ~ 0°C by adding equal volumes of quenching buffer
- The sample is injected in the loop into the pepsin column, digested on column, then peptides are trapped and desalted by LC. All is performed with controlled temperature of
- peptides are eluted by another capillary LC over one-hour gradient water / acetonitrile, electro-sprayed and analysed by MS/MS

HDX-MS measurement of both free and drug-bound forms:

- when the feasibility is concluded and optimized, with a protein coverage of at least 75% the full experiment is repeated according to such conditions with both free and ligand-bound protein
- The ligand-bound protein is produced with incubation of the protein at 4°C overnight and 3-10 molar excess of the ligand (drug), depending on the predicted or measured affinity values, or with 10 x molar excess. For small molecule drugs the excess can be higher, for antibodies it is advised not to oversaturate as this will decrease the digestion efficiency on the protein analyte for HDX-MS
- The full experiment is then carried out automatically for all time points and replicates of the free and bound protein forms.

Data analysis of single and integrated proteomics strategies

MS/MS acquired data are searched against the proper database for peptide identification using different software indicated in the section 4.2.3 above.

Then the results and the spectra are loaded on the software HDExaminer 3.1 where the data is processed, generating the deuterium incorporation of both bound and free protein at different time points and calculating the differential values and heatmaps. The software considers many overlapping peptides cut by pepsin and calculates the minimal region of exchange (a few amino acids) in the sequence, producing a heatmap of the sequence with a differential deuterium incorporation. Such information can also be visualized over the PDB model

structure where the regions identified in the sequence can enlight structural meaning of the drug-binding effects.

4.3 Information for consumables/ chemicals/ equipment

The full pipeline with connected workflows from samples to data analysis requires fully equipped laboratories for:

- cell culture and cell culture treatment, with one lab for mammalian and another one for bacterial cells, equipped with BLS2 cabinets and incubators, centrifuges, microscopes, cell counter, etc.;
- 80°C, -20°C or liquid nitrogen freezers;
- any type of biological sample processing and extraction (lysates, intact cells, tissues, or bodily fluids), with probe and bath sonication systems, homogenizers, thermocyclers and thermal shakers, UV-Vis readers, speed-vacuum systems, centrifuges and an ultracentrifuge (Beckmann XPN-80);
- sample processing for any proteomics workflow and sample preparation for LC-MS analysis, including systems for desalting and buffer exchange, capillary liquid chromatography (LC) for small-scale protein purification, peptide separation and fractionation, as needed as semi-orthogonal LC dimension prior the identification and quantification of peptides/proteins in a proteome-wide scale by LC-MS/MS;
- nanoLC-MS/MS for proteomics, with two high resolution MS systems connected to nanoLCs. The MS instrument park currently includes ten MS instruments, listed as follows: a 2022 Tribrid Orbitrap Eclipse with ETD; three Thermo Orbitrap Exploris 480 (2020-2021); a 2019 tribrid Orbitrap Fusion Lumos; a 2019 tribrid Orbitrap Fusion Lumos with ETD; two Thermo Q-Exactive-HF (2018 and 2019); a 2015 Orbitrap Fusion with ETD; and a 2014 Q-Exactive.
- the most modern platform in Sweden for highly controlled HDX-MS, including micro-LC, a HDX Trajan Parallel liquid handling system and high resolution MS;
- computing instrumentation for database searches on the acquired LC-MS raw files, for chemical proteomics and HDX-MS, and data analysis and visualization with various software.

5 Further applicable documents

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